ECON485 Fall 2025 – Homework Assignment 3

Course Registration System – SQL Homework (Part II)

This homework continues from Homework Assignments 1 and 2. In Homework 1, you designed a 2NF database for a simple course registration system. In Homework 2, you implemented this design in SQL and introduced prerequisite-related structures. In this homework, you will write SQL queries using JOIN operations to produce useful reports and validation checks for the same system.

Please use your Homework 1 and Homework 2 designs as the reference point for this homework. You may add missing example data as needed.

Unless otherwise stated, the database does not need to enforce any business logic automatically. Your goal is to produce meaningful SQL queries that retrieve information from your existing tables. You may use AI tools for assistance, but you must follow the AI usage reporting rules described in the syllabus.

Part 1 – Basic JOIN Queries

In this section, you will write introductory JOIN queries that retrieve common information from the course registration system. Use your own table names and field names. The examples below assume that you have Students, Courses, Sections, Registrations, Prerequisites, and Completed Courses tables, but your names may differ depending on your design.

Task 1 – List Students and Their Registered Sections

Write an SQL query that displays a list of students together with the course code and section number of the sections in which they are registered.

Your query should include at least the following columns (in the output):

- Student name
- Course code
- Section number

Use appropriate JOIN statements across your student, registration, section, and course tables.

Task 2 – Show Courses with Total Student Counts

Write an SQL query that lists all courses together with the total number of students registered in all of their sections.

Your query should:

- JOIN Courses, Sections, and Registrations
- Use GROUP BY to count students
- Display courses with zero registrations as well (Hint: Use LEFT JOIN to include courses without registrations.)

Task 3 – List All Prerequisites for Each Course

Write an SQL query that lists every course together with its prerequisite courses and the minimum grade required to satisfy each prerequisite.

Your query should JOIN:

- The course table
- The prerequisite table
- The table containing prerequisite course information (or the course table again if you used a self-reference)

Part 2 – Intermediate JOIN Query

Task 4 – Identify Students Eligible to Take a Course

Write an SQL query that identifies students who are eligible to register for a given course. A student is eligible only if:

- They have completed all prerequisite courses for that course.
- Their grade for each prerequisite meets or exceeds the minimum required grade.
- They are not already registered for the course.

Your output should include:

- Student name
- Target course
- Each prerequisite course
- Student’s grade for that prerequisite
- A clear eligibility assessment based on the above criteria

Your query must use JOIN operations between multiple tables (students, completed courses, prerequisites, registrations, etc.). You may use **subqueries or aggregation** if required.

Part 3 – Advanced JOIN Query

Task 5 – Detect Students Who Registered Without Meeting Prerequisites

Write one or more SQL statements that identify students who registered for a course without satisfying its prerequisites. A prerequisite violation occurs when:

- A student registers for a course that has prerequisite requirements, and
- The student either did not complete the prerequisite course at all, or
- Completed it with a grade below the minimum required grade.

Your query (or queries) should return at least the following columns:

- Student name
- Course code
- Missing or failed prerequisite course
- Student's grade for that prerequisite (if available)
- Minimum required grade

This task will require multi-table JOIN operations and careful handling of missing records (for example, via LEFT JOIN). Any correct and logically sound SQL approach is acceptable.

Submission Requirements

Your submission must include:

- A single SQL file (hw3.sql) containing all five queries.
- Output for each query, either as text or image capture.
- A README file explaining any assumptions made and documenting your AI usage, if applicable.